

Trade Study TS-002

Ground Control Station Computer Architecture Selection

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

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1. Objective

The objective of this trade study is to evaluate candidate computing architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the solution that best satisfies the project's operational, technical, and cost requirements. The Ground Control Station serves as the primary operator interface for mission planning, command and control, telemetry monitoring, payload management, and situational awareness throughout ISR operations.

Because the Ground Control Station functions as a critical subsystem of the TOMTOM architecture, the selected computing platform must support existing mission software while providing sufficient processing capability for future system expansion. The selected architecture must remain compatible with Linux-based operating systems, QGroundControl, ATAK/OpenTAKServer integration, telemetry processing, object detection software, and future AI-enabled mission capabilities.

In addition to meeting functional requirements, the selected computing platform should provide an appropriate balance between processing performance, expandability, maintainability, portability, cost, and long-term sustainability while maximizing reuse of available hardware resources.

2. Design Requirements

Requirement	Value
Operating System	Linux Compatible
Command & Control	QGroundControl Support
Video Processing	Required
Telemetry Processing	Required
AI Expansion	Desired
Portable	Required
Expandable	Required
Budget	<\$500 Preferred
Repairable	Desired

3. Alternatives Considered

Alternative A

Repurposed ASUS G73S Laptop

The ASUS G73S is a legacy gaming laptop that was selected for evaluation because it was already available to the project and possessed computational capabilities suitable for Ground Control Station operations. Rather than operating as a conventional laptop, the internal hardware is being removed from its original chassis and integrated into a custom portable instrument case to create a dedicated field computer. This approach maximizes

the value of existing hardware while providing a modular platform capable of supporting mission software, future hardware expansion, and subsystem integration.

Advantages

- Existing hardware available at no acquisition cost
- High-performance Intel processor
- Dedicated NVIDIA graphics processor
- Large display support
- Easily repairable and upgradeable
- Compatible with Linux operating systems
- Supports existing GCS software

Disadvantages

- Older hardware architecture
- Larger physical footprint
- Higher power consumption than embedded computers

Alternative B

Intel NUC Mini PC

The Intel NUC is a compact desktop computing platform designed to provide desktop-class processing performance in a significantly smaller package than a conventional computer. The platform offers modern processor architectures and excellent Linux compatibility while maintaining relatively low power consumption. Because the NUC lacks an integrated display, keyboard, and battery system, additional peripherals would be required to create a complete Ground Control Station.

Advantages

- Small footprint
- Modern processor architecture
- Good Linux compatibility
- Low power consumption

Disadvantages

- Additional monitor required
- Higher acquisition cost
- Reduced internal expandability

Alternative C

Rugged Tablet Computer

Rugged tablet computers are designed specifically for harsh operating environments and are commonly employed in military and industrial applications. These systems integrate the display, processor, and battery into a single portable device while providing improved environmental protection. Although highly portable, rugged tablets typically provide limited opportunities for internal hardware expansion and customization.

Advantages

- Lightweight
- Integrated display
- Portable
- Battery powered

Disadvantages

- Limited upgrade capability
- Lower processing performance
- Higher acquisition cost
- Difficult to repair

Alternative D

Raspberry Pi 5

The Raspberry Pi 5 is a compact single-board computer intended for embedded computing applications. Its low cost, excellent Linux support, and extensive developer community make it attractive for many robotics and automation projects. However, the platform provides substantially lower processing capability than desktop-class hardware and would require additional peripherals to function as a complete Ground Control Station.

Advantages

- Lowest cost
- Very low power consumption
- Compact
- Large developer community

Disadvantages

- Limited processing capability
- Not ideal for multiple concurrent applications
- Reduced AI capability
- External peripherals required

4. Evaluation Criteria

Criterion	Weight
Cost	20%
Processing Performance	25%
Expandability	15%
Linux Compatibility	10%
Portability	10%
Maintainability	10%
Future AI Capability	10%

The candidate computing architectures were evaluated using a weighted decision matrix based on technical performance, lifecycle considerations, and project constraints established for the TOMTOM Ground Control Station. Evaluation criteria were selected to reflect the operational requirements identified during system design while balancing technical capability against cost and long-term maintainability. Weighting factors were assigned according to the relative importance of each criterion to the overall mission objectives.

Processing performance received the highest weighting because the Ground Control Station must simultaneously support mission planning, command and control, telemetry monitoring, video processing, and future AI-assisted applications. Cost was also assigned a significant weighting to ensure the selected architecture remained compatible with the project's available resources. Expandability and maintainability were included to support future subsystem integration and long-term development, while Linux compatibility, portability, and future AI capability were evaluated to ensure compatibility with the planned software architecture and anticipated system growth.

5. Decision Matrix

Criteria	Weight	ASUS G73S	Intel NUC	Rugged Tablet	Raspberry Pi 5
Cost	0.20	5	2	2	5
Processing Performance	0.25	5	4	3	2
Expandability	0.15	5	4	2	3
Linux Compatibility	0.10	5	5	4	5

Portability	0.10	3	5	5	5
Maintainability	0.10	5	4	2	4
Future AI Capability	0.10	4	5	2	2

Each alternative was evaluated using a qualitative scoring scale ranging from one to five, where a score of one represents poor performance relative to the evaluation criterion and a score of five represents excellent performance. Individual criterion scores were assigned based on engineering judgment, published hardware specifications, software compatibility, anticipated integration effort, and the operational requirements established for the TOMTOM Ground Control Station.

The weighted decision matrix provides a structured and repeatable method for comparing multiple alternatives while reducing subjective bias during the design process. By multiplying each criterion score by its associated weighting factor, the decision matrix emphasizes characteristics considered most critical to successful system performance while allowing lower-priority considerations to influence, but not dominate, the final engineering decision.

6. Weighted Scores

Alternative	Total Score
Repurposed ASUS G73S	4.70
Intel NUC	3.75
Rugged Tablet	2.75
Raspberry Pi 5	3.60

Weighted scores were calculated by multiplying the score assigned to each evaluation criterion by its corresponding weighting factor and summing the resulting values for each candidate architecture. The resulting total score represents the overall suitability of each alternative relative to the established project requirements.

Although the weighted score serves as the primary decision metric, the final engineering decision also considers practical implementation factors, including hardware availability, subsystem compatibility, future expansion potential, and overall project objectives. Consequently, the selected alternative represents the solution that provides the best overall balance between technical capability, lifecycle value, and implementation risk rather than simply the highest numerical score.

7. Discussion

The repurposed ASUS G73S laptop achieved the highest weighted score among the evaluated alternatives and was selected as the preferred computing architecture for the

TOMTOM Ground Control Station. Although the platform is based on older hardware, it provides the best overall balance of processing performance, expandability, maintainability, and cost while supporting the operational objectives of the project.

A significant advantage of the selected architecture is the availability of existing hardware, eliminating the need to purchase an additional computing platform while providing sufficient computational capability for command and control software, telemetry monitoring, video processing, and future AI-assisted mission functions. The modular design of the laptop also enables replacement of storage devices, memory, cooling components, and other internal hardware as the system evolves.

While newer platforms such as the Intel NUC offer improved energy efficiency and smaller physical dimensions, the additional acquisition cost and requirement for external peripherals reduce their overall value within the project's current constraints. Likewise, embedded computing platforms such as the Raspberry Pi 5 provide excellent power efficiency but lack the processing capability necessary for reliable execution of multiple concurrent mission applications.

The selected architecture supports future integration of companion computers, additional communications hardware, AI accelerators, and upgraded storage devices while preserving compatibility with the Linux software ecosystem established for the TOMTOM Ground Control Station.

8. Decision

Selected Alternative: Repurposed ASUS G73S Laptop

Rationale

- Processing performance
- Cost
- Expandability
- Linux compatibility
- Maintainability
- Future system growth

The selected architecture maximizes the use of existing hardware while providing sufficient computational capability for current mission requirements and future expansion of the TOMTOM Ground Control Station. Future upgrades may incorporate dedicated AI accelerators or additional companion computing hardware without requiring replacement of the primary Ground Control Station architecture.